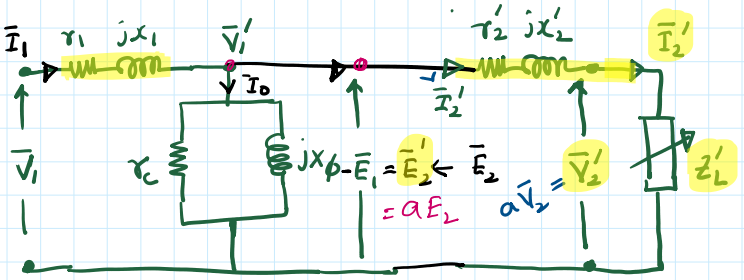
Lecture 38: Transformer Parameters using Open Circuit and Short Circuit Tests



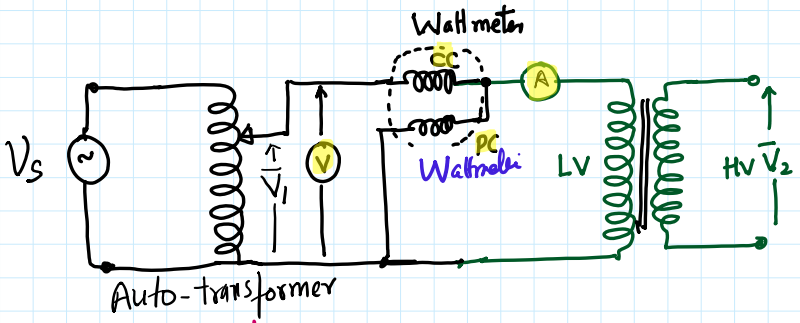
Parameter determination of transformer

Two tests are conducted on transformer to determine its circuit parameters.

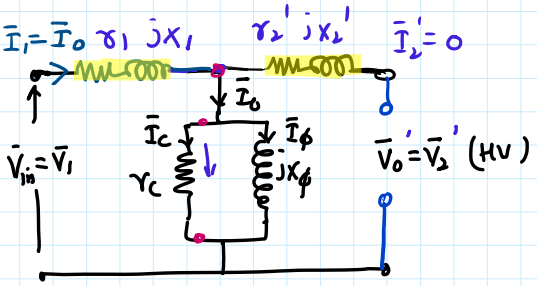
1. Open Circuit Test (OC test): (on LV side) — Open circuit test is conducted by opening secondary @ High voltage.
2. Short Circuit Test (SC test): (on HV side) — Short circuit test is conducted by shorting secondary @ Low voltage. Approximate equivalent circuit at no load.

LV = low voltage
HV = High voltage

Open Circuit Test: is conducted on LV side. The secondary winding is on high voltage side (HV side). A rated voltage is applied at the input to induce rated voltage at the secondary as shown below.

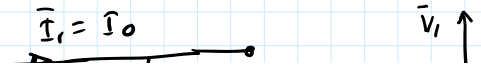


Connection diagram at open circuit test (or no load test)



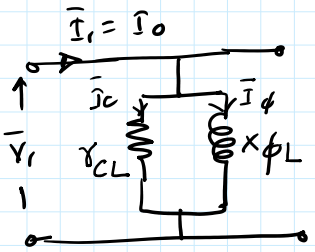
Wattmeter: ① PC: Pressure Coil or voltage coil
② CC: Current coil

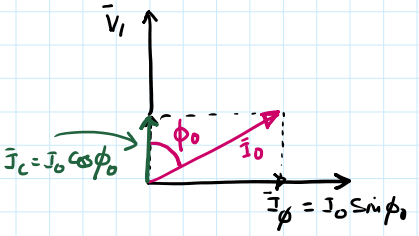
Wattmeter reading: W



Wattmeter reading: W
Voltmeter reading: V_1
Ammeter reading : $I_1 = I_0$

Since current, $\bar{I}_1$ is very small, drop $\bar{V}_{z_1} = \bar{I}_1 z_1$ is neglected.





$$W = P = I_c^2 * r_{CL}$$

$$r_{CL} = \frac{W}{I_c^2}$$

$$I_c = I_0 \cos\phi_0$$

$$I_\phi = I_0 \sin\phi_0$$

$$P = V_1 I_1 \cos\phi_0$$

$$\Rightarrow \cos\phi_0 = \left(\frac{P}{V_1 I_1}\right) = \frac{P}{V_1 I_0}$$

$\sin\phi_0$

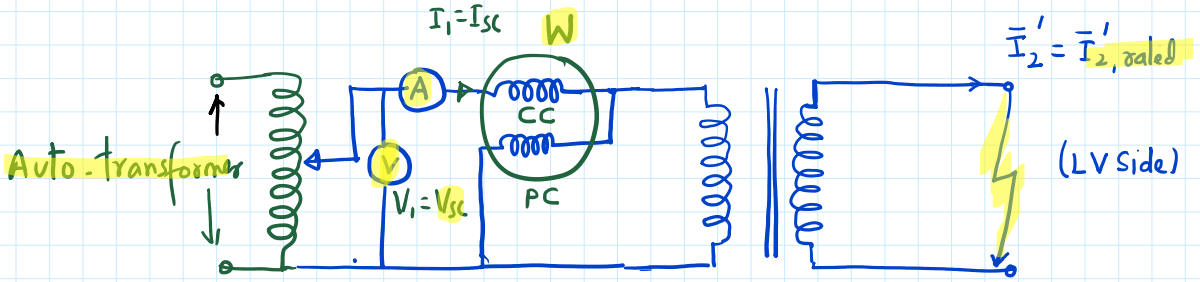
$$r_{CL} = \frac{V_1}{I_c} = \frac{W}{I_c^2}$$

$$X_{\phi L} = \frac{V_1}{I_\phi} = \frac{V_1}{I_0 \sin\phi_0}$$

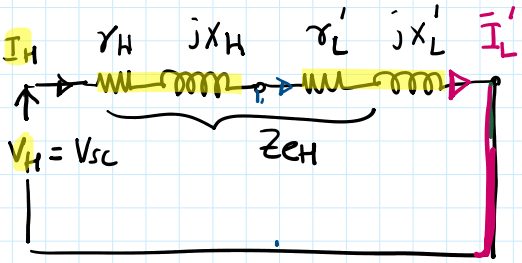
$$r'_{CL} = r_{CL} * \left(\frac{N_H}{N_L}\right)^2 = r_{CH}$$

$$X'_{\phi L} = X_{\phi L}\left(\frac{N_H}{N_L}\right)^2 = X_{\phi H}$$

Short Circuit Test:



Wattmeter: W
Voltmeter : V_{SC}
Ammeter : I_{SC}



$$|Z_{eH}| = \frac{V_{SC}}{I_{SC}} = \sqrt{r_{eH}^2 + X_{eH}^2} \quad -(1)$$

$$r_{eH} = r_H + r'_L$$

$$X_{eH} = X_H + X'_L$$

$$Z_{eH} = (r_H + r'_L) + j(X_H + X'_L)$$

$$W = P = I_H^2 * r_{eH} = I_{SC}^2 * r_{eH}$$

$$r_{eH} = \frac{P}{\quad} \quad -(2)$$

$W = P =$ $I_H^2 r_{eH}$ ...

$$r_{eH} = \frac{P}{I_{sc}^2} \quad \text{---(2)}$$

$$Z_{eH} = (r_H + r_L') + j(x_H + x_L')$$

$$x_{eH} = \sqrt{z_{eH}^2 - r_{eH}^2} \quad \text{---(3)}$$

$$r_{eH} = r_H + r_L' \; ; \quad r_H \approx r_L'$$

$$x_H \approx x_L'$$

$$\Rightarrow r_H \approx r_L' = \frac{r_{eH}}{2}$$

$$x_H \approx x_L' = \frac{x_{eH}}{2}$$

Example: A 20KVA, 2500/250 V, 50 Hz, 1-ϕ transformer gave the following test results.

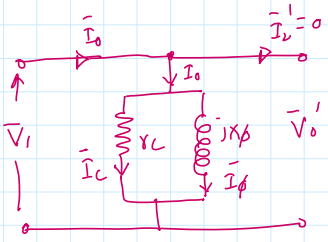
	V	I	W
Open circuit test (on LV side):	250V,	1.4A,	105 W
Short Circuit test (on HV side)	104V,	8A,	320W

Compute the parameters of approximate equivalent circuit referred to HV and LV side.

Solution: Open Circuit Test:

$$W = P_c = V_1 I_0 \cos\theta_0 \Rightarrow \cos\theta_0 = \frac{P_c}{V_1 I_0} = \frac{105}{250 \times 1.4} = 0.3$$

$$\theta_0 = \cos^{-1}(0.3) = 72.54^\circ$$



$$\Rightarrow I_c = I_0 \cos\theta_0 = 1.4 \times \cos 72.54 = 0.42\,A$$

$$I_\phi = I_0 \sin\theta_0 = 1.4 \times \sin 72.54 = 1.335\,A$$

$$\Rightarrow r_c = \frac{V_1}{I_c} = \frac{250}{0.42} = 595.24\,\Omega$$

$$x_\phi = \frac{V_1}{I_\phi} = \frac{250}{1.335} = 187.265\,\Omega$$

Thus

$r_{cL} = 595.24\,\Omega$

$x_{\phi L} = 187.265\,\Omega$

referred to Low voltage Side

If we want to find high voltage side just multiply by $\left(\frac{N_2}{N_1}\right)^2 = \left(\frac{N_H}{N_L}\right)^2$, Here $n = \frac{2500}{250} = 10$

$n^2 = 100$

$\left(\frac{N_L}{N_1}\right) = \left(\frac{N_H}{N_L}\right)$, Here $n = \frac{250}{\cancel{250}}$

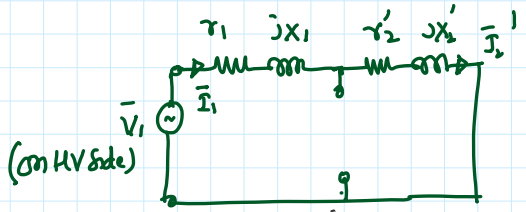
$n^2 = 100$

Core losses
Iron losses
Fixed losses

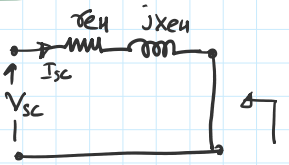
$\rightarrow r'_{CL} = r_{CH} = \left(\frac{N_H}{N_L}\right)^2 * r_{CL} = 59524\,\Omega$

$x'_{\phi L} = x_{\phi H} = \left(\frac{N_H}{N_L}\right)^2 * x_{\phi L} = 18726.5\,\Omega$

Short Circuit Test: (HV side)



(on HV side)



$r_{eH} = r_H + r'_L$

$x_{eH} = x_H + x'_L$

$z_{eH} = r_{eH} + jx_{eH}$

$|z_{eH}| = \frac{V_{sc}}{I_{sc}} = \frac{104}{8} = 13\,\Omega$

$r_{eH} = \frac{P_{sc}}{I_{sc}^2} = \frac{320}{8^2} = 5\,\Omega$

$x_{eH} = \sqrt{|z_{eH}|^2 - r_{eH}^2} = \sqrt{13^2 - 5^2} = 12\,\Omega$

$r_{eH} = 5\,\Omega = r_H + r'_L$ } referred to HV side.

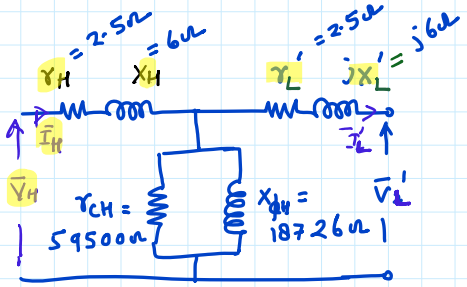
$x_{eH} = 12\,\Omega = x_H + x'_L$

Normally $r_H \simeq r'_L = \frac{r_{eH}}{2} = 2.5\,\Omega$

$x_H \simeq x'_L = \frac{x_{eH}}{2} = 6\,\Omega$

$r_{CH} = 59524\,\Omega$

$x_{\phi H} = 18726.5\,\Omega$



Trans. equivalent ckt referred to HV side.

$r_{eL} = \left(\frac{N_L}{N_H}\right)^2 \times 5 = \frac{5}{10^2} = 0.05\,\Omega$

$x_{eH} = \left(\frac{N_L}{N_H}\right)^2 * 12 = \frac{12}{100} = 0.12\,\Omega$

} referred to LV side

$\Rightarrow r_{eL} = r_{1L} + r'_{2H} \Rightarrow r_{1L} \simeq r'_{2H} = \frac{0.05}{2} = 0.025\,\Omega$

$x_{eL} = x_{1L} + x'_{2H}$ $\quad x_{1L} \simeq x'_{2H} = \frac{0.12}{2} = 0.06\,\Omega$

$r_{CL} = 595.24\,\Omega$

$x_{\phi L} = 187.265\,\Omega$

